Computer Science Department
Sofware Engineering & Business

Bachelor's Thesis

MovieCrush

Movie Tracking Platform with Recommendations, Detailed Rati

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TODOS

1.4 Boxes 10

- Personnal todo before marking this thesis a

Academic Integrity Statement

I, undersigned, hereby declare that this caps

- All ideas, data, fgures and text from other
  in the bibliography.
- No part of this project has been submitted p
  any other institution.
- All code, diagrams, and third-party material
  with permission and properly referenced.
- I have not engaged in plagiarism or any form
- Any assistance received (e.g. from peers, tu
  the acknowledgements section.

I understand that failure to comply with thes
misconduct and may lead to disciplinary actio

Place, date Kyiv, 02.06.2026

Signature _____

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I want to thank my grandparents - they were a and were even willing to learn how to code ju for the fact that when I had classes until 9 minutes instead of an hour and a half, and th well every time I came over, as if it was a w want to thank my brother for always showing u a student here at that moment, just to help o

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the best it could be - in such difcult times.

1 Introduction

Your introduction serves to introduce the topic of your Bachelor thesis and to arouse the reader's curiosity with an overview. Why it is important and how it is structured, we explain here.

You can consider an introduction as a teaser for your bachelor thesis. You arouse interest and give a foretaste by presenting your motivation, your method and the state of research in your introduction.

Convince your examiners already in the introduction that your Bachelor thesis will be exciting. If your professor starts reading your thesis with anticipation and interest, the chances of getting good grades are higher.

Pay particular attention to the following in your introduction:

- ***Introduce the topic*** *- What characterizes the topic?*
- ***Introduce the goal*** *- What do you want to achieve with your thesis?*
- ***Make the reader curious*** *- What motivates the reader to read on?*
- ***Describe the relevance*** *- Why is this bachelor thesis scientifically relevant?*

The introduction should have the following content:

- ***Initial situation presentation of the topic*** *- You introduce the topic with an exciting 'bait'. You provide initial information on the topic and the object of research and explain the current state of research.*
- ***Relevance of the topic motivation*** *- You justify the relevance of your topic (scientifically) and place it in the context of your field. In addition, it is often required that you disclose your personal motivation.*
- ***Objectives*** *- Your introduction should clearly state what the goal of your paper is and what outcome you hope to achieve upon completion of the bachelor thesis.*
- ***Method*** *- You explain the approach and justify the choice of method.*
- ***Structure of the Bachelor's thesis*** *- Finally, you give the reader a general overview of your Bachelor's thesis by explaining the structure, showing the red thread and how the research question is answered.*

> Welcome to the template's introductory
> you with lorem ipsum, here's a quick gu
> Typst and, more specifcally, in this ter
> Need more? Check out Typst

1.1 Basic markup

Typst lets you create bold, italic, or monosp
in equations $e^(i pi) + 1 = 0$ or even in
```
fn main() { println!("Hello,
World!"); }
```
. And because I prefer to use British
more Boldly colorize!

You can also write numbered or unnumbered lis
- First item
- Second item
  1. First Subitem

2 Second Subitem
- Third item

Need equations? Sure! They look great as bloc

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

1.2 Images

As they say, a picture is worth a thousand wo

Figure 2: Project logo

1.3 Tables

Tables are great for organizing data. From si

Name	Age	City
Albert Einstein	25	Berlin
Marie Curie	22	Paris
Isaac Newton	30	London

Table 1: Simple table

[31:27]	[24:20]	[19:15]	[14:12]	[11:7]	[6:0]
funct5	aq	rl	rs2	rs1	funct3	opcode
5		5	5	5	3	5	7

Table 2: Complex table

1.4 Boxes

Highlight key points with these fun boxes (an

Infobox: - For lighting info

Ideabox: - Share a bri liant idea.

Warningbox: P
with caution!

Firebox! This is

Rocketbox: Sh
the moon!

Todobox: Just do it!

TODO

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1.5 Citations, Acronyms and Glossary

Add citations in `/tail/bibliography.bib`.

Acronyms are defined in `/tail/glossary.typ`.

1.6 Code

Besides writing `fn main() { println!("Hello World"); }` you can also write code blocks like this:

```rust
fn main() {
  let ship = Starship::new("USS Rustacean", (0.0, 0.0, 0.0));
  let destination = (42.0, 13.0, 7.0);
  let warp = ship.optimal_warp(ship.distance_to(destination));

  println!("🖖 {} traveling to {:?} at Warp {:.2}", ship.name, destination, warp);
  if warp <= 9.0 {
    println!("✨ Warp engaged!");
  } else {
    println!("⚠ Warp failed!");
  }
}
```

Listing 1: First part of the USS-Rust

or directly from a fle

```rust
struct Starship {
  name: String,
  position: (f64, f64, f64),
}

impl Starship {
  fn new(name: &str, position: (f64, f64, f64)) -> Self {
    Self {
      name: name.into(),
      position,
    }
  }
  fn distance_to(&self, dest: (f64, f64, f64)) -> f64 {
    ((dest.0 - self.position.0).powi(2)
      + (dest.1 - self.position.1).powi(2)
      + (dest.2 - self.position.2).powi(2))
    .sqrt()
  }
  fn optimal_warp(&self, distance: f64) -> f64 {
    (distance / 10.0).sqrt().min(9.0)
  }
}
```

Listing 2: Second part of `/resources/code/uss-rustacean.rs`

1.7 Context Problem

Haute École d'Ingénierie et de Gestion du Canton de Vaud (HEIG-VD)

[1, 1, p. 7f]

```rust
fn main() {
  println!("Hello World!");
}
```

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1.8 Objectives

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1.9 Structure of this report

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2.1 Section 1

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2.2 Section 2

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2.3 Conclusion

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3.1 Section 1

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4.1 Section 1

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5 Validation

Validation (Requirements Verification and Testing)

In this section you demonstrate how your implementation satisfies the initial requirements through clear testing methods and concise examples—suitable for a bachelor-level project:

1. *Restate each key functional and non-functional requirement from your Analysis and Design sections*
2. *Describe the testing approach for each requirement (for example, unit tests, manual acceptance checks or scenario walkthroughs)*
3. *Provide concrete test cases or usage examples that show how you verify each requirement in practice*
4. *Summarize actual versus expected outcomes, indicating pass/fail status for each test*
5. *Include brief snippets of test code or sample console outputs to illustrate your procedures*
6. *Note any gaps or deviations and suggest simple fixes or areas for future improvement*
7. *If a feature wasn't intended for specific scenarios (e.g. high-load), omit unrealistic stress tests and clearly document its current limitations*

This focused structure ties every requirement directly to validation results, using examples and methods you can realistically carry out at the bachelor level.

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Contents

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5.1 Section 1

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5.2 Section 2

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5.3 Conclusion

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Glossary

Programming Language

Rust **– Rust Programming Language** : Rust is a modern systems pro
focused on safety, speed, and concurrency. I
such as null pointer dereferencing and data
choice for performance-critical applications
6, 7

University

HEI **– Haute École d'Ingénierie** 7

IT **– Infotronics** 6

Bibliography

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turing Technology, vol. 3, May 2023, doi:
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